

Phase 2 Insight Report

Social Engine Recovery - SQL Analytics

Data Vortex :: AARUUSH'26 | Team SPARTANS

Executive Summary

After restoring the Social Engine dataset in Phase 1, we loaded the cleaned data into a normalized SQLite database and executed 16 analytical SQL queries across Easy, Medium, and Hard difficulty tiers. Our analysis uncovered critical insights about platform engagement patterns, user behaviour anomalies, and data corruption profiles. Below are our key findings.

1. Platform Landscape is Balanced

All 5 platforms (Facebook, YouTube, Twitter, Reddit, Instagram) have nearly equal post volumes, within 4% of each other. Facebook leads by a slim margin (2,074 posts). This balanced distribution suggests the Social Engine had equal platform integration before the crash.

2. YouTube Drives Highest Per-Post Engagement

Despite Facebook having the most posts, YouTube generates the highest average total engagement at 4,044.54 per post. This confirms that video-based content consistently outperforms text-only platforms in driving user interaction.

3. The Follower Count Paradox

Our M2 analysis reveals a counter-intuitive finding: users with fewer than 25K followers actually achieve slightly higher average engagement (3,535.84) compared to high-follower users (3,508.45). This challenges the common assumption that more followers equals more engagement and suggests content quality matters more than audience size.

4. Geographic Engagement Hotspots

Los Angeles generates the most total engagement (1.64M), followed by Munich (1.62M) and Shanghai (1.59M). These three cities represent the Social Engine's most active user bases. Interestingly, the top 10 locations are geographically diverse, spanning 4 continents.

5. Massive Suspicious Sharing Activity

24.3% of all posts (2,915) exhibit suspicious sharing patterns where shares exceed the combined total of likes and comments. This unusually high ratio may indicate bot-driven amplification, share-farming networks, or artificial engagement inflation. This requires further investigation.

6. Uniformly Distributed Engagement (Anomaly)

No single user's average engagement exceeds 2x the overall average (threshold: 7,044.63). In real social media data, we would expect power-law distributions with viral outliers. This uniformity suggests the data may have been synthetically generated or heavily smoothed.

7. Low-Follower High-Impact Anomalies

14 users with fewer than 5,000 followers rank in the top 10% of total engagement. The most anomalous is user_uerv85na from Rome (1,824 followers, 60,629 total engagement across 16 posts). These accounts may represent: organic viral creators, company-operated accounts, or engagement manipulation.

8. Data Corruption Impact Assessment

The corrupted dataset contained 4,274 anomalous posts (34.5%): 525 with negative likes (impossible values), 1,846 with missing platforms, 1,746 with missing text content, and 663 containing HTML artifacts. Missing platform identifiers were the most common corruption pattern.

9. The Most Suspicious User: user_uerv85na

This user appears as an anomaly across THREE separate hard-level queries (H4, H6, and implicitly H1). With only 1,824 followers but 60,629 total engagement, an average of 3,789 per post, and posts where shares exceed likes, this account represents the single most suspicious entity in the entire dataset.

10. Instagram is Best for Influencer Marketing

Among users with 30K+ followers, Instagram delivers the highest average engagement per post (3,622.78), outperforming Reddit (3,523.90), Twitter (3,491.04), YouTube (3,488.44), and Facebook (3,441.39). For influencer marketing campaigns, Instagram should be the primary channel.

Conclusion

Our SQL analysis of the recovered Social Engine dataset reveals a platform with balanced cross-platform presence but significant engagement anomalies. The most concerning finding is the 24.3% suspicious sharing rate, combined with 120 users who meet multiple suspicion criteria. The uniformly distributed engagement pattern (no user exceeds 2x average) is itself an anomaly that warrants investigation. We recommend the Social Engine team implement: (1) Bot detection algorithms targeting high share-to-like ratios, (2) Follower-to-engagement audit systems, and (3) Platform-specific engagement normalization to account for content format differences.